

Validated Peripheral Rank-Two Deflation for Repaired Folded-Gaussian Matrices

Stationary Neumann Solves, Bordered Parity Newton Balls, and Certified Riesz
Projectors

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Abstract

RH-72 replaced the rounded folded-Gaussian rows by exact nonnegative dyadic stochastic rows and certified their assembly error. The Perron right vector is therefore exactly $\mathbf{1}$, but the stationary left vector and the negative parity Riesz factor remained unvalidated. This paper closes that finite-scale spectral gate.

For an exact row-stochastic matrix $A \in \mathbb{Q}^{n \times n}$, define

$$L = I - A^T + \frac{1}{n} \mathbf{1} \mathbf{1}^T.$$

If an approximate inverse B obeys $\|I - BL\|_F < 1$, then L is invertible. A single pre-conditioned residual encloses the normalized stationary vector and proves that the Perron eigenvalue is simple. The negative parity pair is treated as a zero of

$$F(r, \lambda) = \begin{pmatrix} Ar - \lambda r \\ c^T r - 1 \end{pmatrix}.$$

For a bordered approximate inverse, the computable inequalities

$$\beta + \gamma\rho + M\rho^2 \leq \rho, \quad \gamma + M\rho < 1$$

give a unique joint eigenpair in the radius- ρ ball. A second bordered linear solve validates the left eigenvector at the resulting eigenvalue. We derive an explicit normalized rank-one projector perturbation bound and combine it with the stationary factor to enclose the complete rank-two deflation and the deflated bulk operator. A Euclidean Grushin/Rouche ledger independently proves that a radius-0.01 circle contains exactly one parity eigenvalue.

A 160-bit Arb audit covers both fine and Haar-coarse repaired matrices at $\sigma = 0.16, 0.08, 0.04, 0.02, 0.01$. All ten channels close. The largest stationary, right-pair, and left-vector errors are respectively 2.10×10^{-16} , 3.54×10^{-15} , and 8.22×10^{-14} . The largest rank-two projector and deflated-bulk errors are 9.20×10^{-14} and 9.41×10^{-14} ; the largest contour transport product is 0.241. Gram products, residuals, and final projector bounds are evaluated from exact dyadic data with outward rounding.

Thus the repaired finite matrices now have certified Perron/parity rank-two deflations. The normalized source/observation transfer to RH-70's frozen Hardy input remains the next gate. No small-noise uniform theorem, Stage A1 closure, Hilbert–Polya operator, or Riemann Hypothesis claim is made.

Keywords: stochastic matrix; stationary distribution; validated eigenpair; Riesz projector; interval arithmetic; Grushin problem.

MSC 2020: 47A10; 47A55; 60J20; 65G20; 65F15.

1 Introduction

The RH sequence studies spectral structures generated by a noisy quadratic map at its first band-merging parameter. Earlier calculations isolated a near- -1 parity branch and a complementary bulk, but a production-level certificate must distinguish a floating eigendecomposition from a theorem about the exact matrix being used.

RH-70 validated a terminal block-Hardy calculation after accepting frozen binary64 operator, source, and observation arrays as exact dyadic inputs [7]. RH-71 identified the upstream construction of those arrays as the remaining finite-scale frontier. RH-72 then certified the folded-Gaussian kernel, support truncation, normalization, exact stochastic repair, and Haar assembly [5]. In particular, the repaired matrix satisfies $A\mathbf{1} = \mathbf{1}$ exactly.

The remaining spectral problem is nonnormal. A small residual alone does not prove that an eigenpair exists nearby, nor that the corresponding Riesz projector is well conditioned [1, 3]. We therefore use three separate certificates:

1. a linear Neumann certificate for the stationary left vector;
2. a nonlinear bordered contraction for the right parity eigenpair, followed by a linear interval solve for its left partner;
3. a Grushin/Rouche count on a fixed circle, which records spectral isolation independently of a diagnostic eigengap.

The calculations use binary64 only to propose centers. Every matrix entry and center coordinate is then converted to its exact dyadic rational, and all residuals, products, norms, and inequalities are recomputed in Arb [2, 4].

Contributions

1. We prove and execute a stationary Neumann certificate that also proves simplicity of the Perron eigenvalue.
2. We give a bordered Newton validation for a normalized right eigenpair.
3. We transfer the validated eigenvalue ball through a bordered left solve.
4. We derive a projector perturbation formula adapted to the two different normalizations used by the right and left systems.
5. We enclose the rank-two projector and deflated bulk errors.
6. We close a radius-0.01 parity contour count in every fine/coarse channel at the five archived noise scales.

2 Exact repaired matrices and notation

At each noise scale, RH-72 supplies an exact nonnegative dyadic matrix $A \in \mathbb{Q}^{n \times n}$ with

$$A\mathbf{1} = \mathbf{1}. \tag{1}$$

The fine dimensions are 32, 64, 128, 256, 512. The coarse matrix is formed by the exact rational Haar averaging rule

$$A_{ij}^{(c)} = \frac{1}{2} \sum_{a,b \in \{0,1\}} A_{2i+a, 2j+b}. \tag{2}$$

It is again nonnegative and exactly row-stochastic.

All vector norms are Euclidean and all matrix norms in the proofs are subordinate two-norms. The implementation may replace a two-norm by the Frobenius norm, since $\|X\|_2 \leq \|X\|_F$. An overline such as $\bar{x}$ denotes a rigorously rounded upper bound, not complex conjugation. Transposes may be replaced by adjoints without changing the arguments.

3 Stationary Neumann certificate

Put $e = \mathbf{1}$ and

$$L = I - A^T + \frac{1}{n}ee^T, \quad b = \frac{1}{n}e. \quad (3)$$

Theorem 3.1 (Stationary Neumann certificate). *Let B be any square matrix and suppose*

$$\gamma = \|I - BL\| < 1.$$

Then L is invertible. The vector $\pi = L^{-1}b$ is the unique normalized stationary vector:

$$A^T \pi = \pi, \quad e^T \pi = 1.$$

For any center π_0 , set

$$\beta = \|B(b - L\pi_0)\|.$$

Then

$$\|\pi - \pi_0\| \leq \frac{\beta}{1 - \gamma}. \quad (4)$$

Moreover, the eigenvalue 1 of A is algebraically simple.

Proof. The inequality implies that $BL = I - (I - BL)$ is invertible by the Neumann series. Since B and L are square, both are invertible. Multiplying $L\pi = b$ on the left by e^T and using $Ae = e$ gives $e^T \pi = 1$; substitution then gives $(I - A^T)\pi = 0$.

If $(I - A^T)x = 0$ and $e^T x = 0$, then $Lx = 0$, hence $x = 0$. Thus the stationary eigenspace is one dimensional. Finally, a nonnegative stochastic matrix is power bounded in the induced infinity norm: $\|A^k\|_\infty = 1$. A nontrivial Jordan block at the unit-modulus eigenvalue 1 would make $\|A^k\|$ grow polynomially, so 1 is semisimple and therefore algebraically simple. For the error, write

$$\pi - \pi_0 = (BL)^{-1}B(b - L\pi_0)$$

and use $\|(BL)^{-1}\| \leq (1 - \gamma)^{-1}$. □

Because the exact Perron right vector is e , the approximate Perron projector $e\pi_0^T$ satisfies

$$\|e\pi^T - e\pi_0^T\| \leq \sqrt{n} \frac{\beta}{1 - \gamma}. \quad (5)$$

4 Bordered Newton validation of the parity pair

Let (r_0, λ_0) be a numerical negative real eigenpair center and let c be a fixed numerical left vector scaled so that $c^T r_0$ is close to one. Define

$$F(r, \lambda) = \begin{pmatrix} Ar - \lambda r \\ c^T r - 1 \end{pmatrix}, \quad J_0 = \begin{pmatrix} A - \lambda_0 I & -r_0 \\ c^T & 0 \end{pmatrix}. \quad (6)$$

Theorem 4.1 (Bordered Newton validation). *Let B be an approximate inverse of J_0 , and define*

$$\beta = \|BF(r_0, \lambda_0)\|, \quad \gamma = \|I - BJ_0\|, \quad M = \|B\|.$$

If a radius $\rho > 0$ satisfies

$$\beta + \gamma\rho + M\rho^2 \leq \rho, \quad \gamma + M\rho < 1, \quad (7)$$

then F has a unique zero (r, λ) in the joint Euclidean ball $\|(r - r_0, \lambda - \lambda_0)\| \leq \rho$.

Proof. For $h = (u, \eta)$, the nonlinear remainder of F at the center is $(-\eta u, 0)$, whose norm is at most $\|h\|^2$. The preconditioned fixed point map $h \mapsto h - BF((r_0, \lambda_0) + h)$ therefore maps the radius- ρ ball into itself by the first inequality in (7). Its Lipschitz constant there is at most $\gamma + M\rho < 1$. Banach's fixed point theorem proves existence and uniqueness. $\square$

The implementation takes Frobenius upper bounds for M and γ . It first proposes $2\beta/(1-\gamma)$ and then checks both inequalities in Arb; no floating decision is accepted without this final interval check.

5 Left parity validation

The right Newton normalization gives $c^T r = 1$. To construct the left mode, fix the right center r_0 and solve, at the unknown validated eigenvalue,

$$H(\lambda) \begin{pmatrix} \ell \\ \alpha \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad H(\lambda) = \begin{pmatrix} A^T - \lambda I & c \\ r_0^T & 0 \end{pmatrix}. \quad (8)$$

Theorem 5.1 (Left parity validation). *Let B_ℓ approximately invert $H(\lambda_0)$ and put*

$$\gamma_\ell = \|I - B_\ell H(\lambda_0)\|, \quad M_\ell = \|B_\ell\|.$$

For a center $x_0 = (\ell_0, 0)$, let

$$\beta_0 = \|B_\ell\{(0, 1)^T - H(\lambda_0)x_0\}\|.$$

If $|\lambda - \lambda_0| \leq \rho$ and

$$\Gamma = \gamma_\ell + M_\ell \rho < 1, \quad (9)$$

then (8) has a unique solution and

$$\|x - x_0\| \leq \frac{\beta_0 + M_\ell \rho \|\ell_0\|}{1 - \Gamma}. \quad (10)$$

For the eigenvalue supplied by Theorem 4.1, its scalar component is $\alpha = 0$; hence $A^T \ell = \lambda \ell$ and $r_0^T \ell = 1$.

Proof. The perturbation $H(\lambda) - H(\lambda_0)$ has norm at most ρ . Preconditioning and a Neumann series give (10). At the validated right eigenpair, multiply the first block equation by r^T :

$$r^T(A^T - \lambda I)\ell + (c^T r)\alpha = 0.$$

The first term vanishes and $c^T r = 1$, so $\alpha = 0$. $\square$

6 Normalized projector and bulk bounds

Write $r = r_0 + \delta r$, $\ell = \ell_0 + \delta \ell$ and suppose

$$\|\delta r\| \leq \varepsilon_r, \text{quad} \|\delta \ell\| \leq \varepsilon_\ell.$$

The left normalization gives $\ell^T r_0 = 1$. Consequently

$$|\ell^T r - 1| = |\ell^T \delta r| \leq (\|\ell_0\| + \varepsilon_\ell) \varepsilon_r =: \eta.$$

Proposition 6.1 (Normalized parity projector). *If $\eta < 1$, then the exact parity Riesz projector $P_- = r \ell^T / (\ell^T r)$ and the center tensor $Q_0 = r_0 \ell_0^T$ obey*

$$\|P_- - Q_0\| \leq \frac{N}{1 - \eta} + \frac{\|r_0\| \|\ell_0\| \eta}{1 - \eta}, \quad (11)$$

where

$$N = \varepsilon_r (\|\ell_0\| + \varepsilon_\ell) + \|r_0\| \varepsilon_\ell.$$

Proof. The numerator difference satisfies

$$\|r \ell^T - r_0 \ell_0^T\| \leq \varepsilon_r (\|\ell_0\| + \varepsilon_\ell) + \|r_0\| \varepsilon_\ell = N.$$

Also $|\ell^T r| \geq 1 - \eta$ and $|1/(\ell^T r) - 1| \leq \eta/(1 - \eta)$. Insert and subtract $r_0 \ell_0^T / (\ell^T r)$. $\square$

Let $P_+ = e \pi^T$, $Q_+ = e \pi_0^T$, and let E_+ and E_- denote the right sides of (5) and (11). Then

$$\|(P_+ + P_-) - (Q_+ + Q_0)\| \leq E_+ + E_-. \quad (12)$$

For the exact and center-deflated bulks

$$K = A - P_+ - \lambda P_-, \quad K_0 = A - Q_+ - \lambda_0 Q_0,$$

we also obtain

$$\|K - K_0\| \leq E_+ + (|\lambda_0| + \rho) E_- + \rho \|r_0\| \|\ell_0\|. \quad (13)$$

7 Parity contour count

The Newton theorem gives a local zero of the normalized eigenproblem. We also certify spectral isolation on a fixed, much larger circle. Let $z = \lambda_0 + w$, $|w| = R$, and use the bordered operator at the center as a Grushin problem. Denote by E_0 a certified upper bound for its reduced inverse and by $g_- = |\ell_0^T r_0|$, $g_+ = |\ell_0^T r_0|$. The RH-42 Euclidean ledger transfers the bordered inverse to the circle whenever

$$R E_0 < 1. \quad (14)$$

It then bounds the right and left channels, the affine scalar boundary, and the residual correction. If the scalar correction is smaller than the affine boundary, Rouché's theorem gives exactly one zero inside the circle [6].

In the present audit $R = 0.01$. The Gram interval is not assumed to equal one: it is recomputed as the exact dyadic dot product of the displayed centers. Likewise both residuals are formed against the exact rational matrix, rather than a rounded dense copy. This is the parity contour count used in the archived gate.

8 Five-scale interval audit

SciPy eigensolvers propose $\pi_0, r_0, \ell_0, \lambda_0$ and approximate inverse centers. Every binary64 number is converted by its exact integer ratio. The repaired fine and coarse matrices are exact fractions. Arb at 160-bit precision then evaluates all matrix residuals, Frobenius norms, Neumann denominators, Newton inequalities, Gram products, projector formulas, and final bulk bounds with outward rounding.

Table 1: Fine/coarse parity centers and certified rank-two errors.

σ	n	$\lambda_-^{(f)}$	$\lambda_-^{(c)}$	$E_2^{(f)}$	$E_2^{(c)}$
0.16	32	-0.86795414	-0.86540565	$3.38 \cdot 10^{-14}$	$1.90 \cdot 10^{-14}$
0.08	64	-0.93787089	-0.93675101	$4.61 \cdot 10^{-14}$	$3.72 \cdot 10^{-14}$
0.04	128	-0.96563294	-0.96505988	$5.00 \cdot 10^{-14}$	$2.75 \cdot 10^{-14}$
0.02	256	-0.97925977	-0.97892302	$5.62 \cdot 10^{-14}$	$4.60 \cdot 10^{-14}$
0.01	512	-0.98665865	-0.98645070	$9.20 \cdot 10^{-14}$	$5.89 \cdot 10^{-14}$

Here E_2 is the complete rank-two projector error from (12). The corresponding maximum deflated-bulk error is 9.41×10^{-14} .

Table 2: Fine-channel validation radii and contour transport products.

σ	stationary error	right radius	left error	$0.01E_0$
0.16	$1.50 \cdot 10^{-16}$	$3.30 \cdot 10^{-15}$	$2.54 \cdot 10^{-14}$	0.0652
0.08	$1.65 \cdot 10^{-16}$	$3.46 \cdot 10^{-15}$	$3.64 \cdot 10^{-14}$	0.0862
0.04	$1.55 \cdot 10^{-16}$	$2.93 \cdot 10^{-15}$	$4.13 \cdot 10^{-14}$	0.1196
0.02	$1.07 \cdot 10^{-16}$	$2.49 \cdot 10^{-15}$	$4.87 \cdot 10^{-14}$	0.1689
0.01	$1.26 \cdot 10^{-16}$	$2.95 \cdot 10^{-15}$	$8.22 \cdot 10^{-14}$	0.2402

Every coarse channel has smaller contour transport than its fine partner. Across all ten channels the smallest exact-dyadic Gram lower bound exceeds 0.9999999999999997, and every Rouché scalar gate is green. The diagnostic distance to the remaining numerical spectrum is archived but is not used as a proof input.

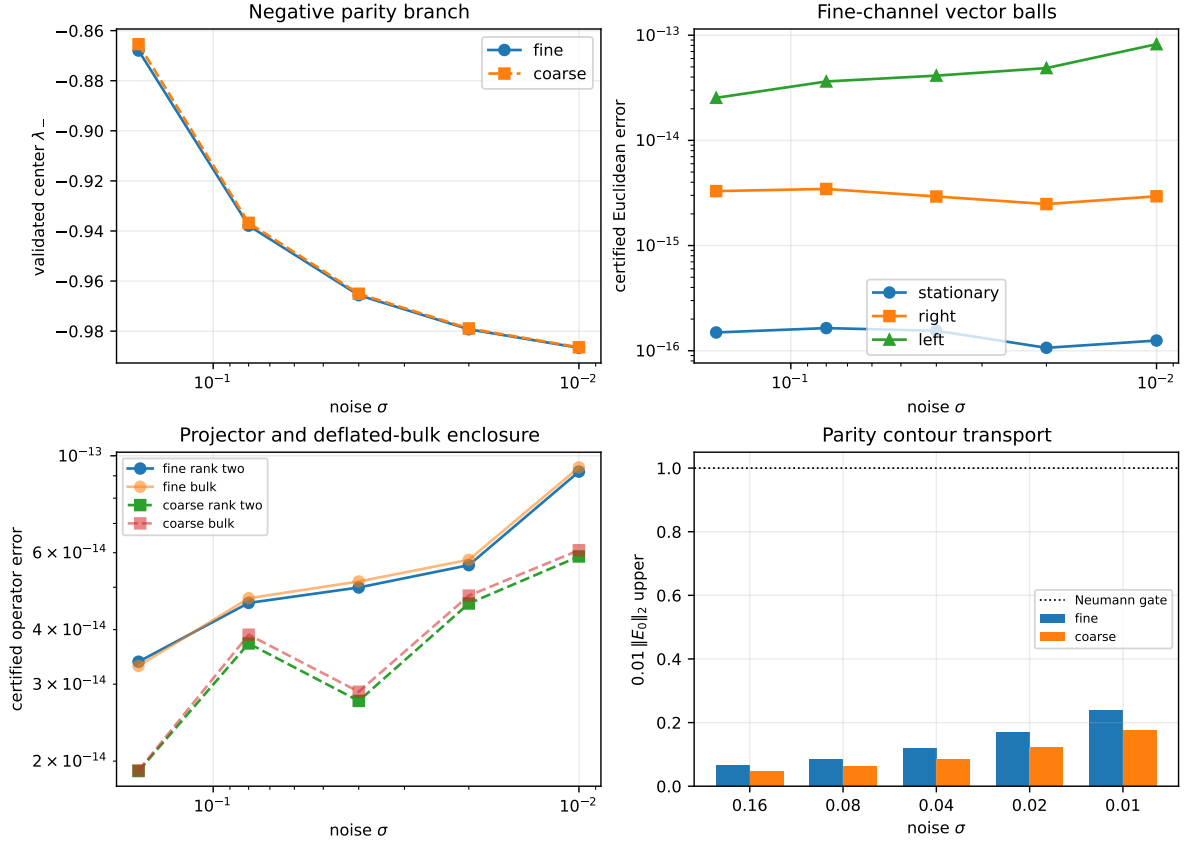


Figure 1: Validated parity branch, vector balls, rank-two/bulk enclosures, and fixed-radius contour transport. Reversing the horizontal noise axis makes the small-noise direction run left to right.

9 What is and is not closed

The result is exact for each archived repaired dyadic matrix. Together with RH-72 it removes the following finite-scale gates:

1. folded-Gaussian and Haar matrix assembly;
2. the exact Perron right vector and validated stationary left vector;
3. the isolated negative parity eigenvalue and both Riesz vectors;
4. the complete rank-two deflation and its center error.

The next task is the *source/observation transfer*: the parity and stationary factor balls must be propagated through all normalizations used to construct the forcing and observation vectors. The resulting upstream operator/source/observation triple must then be compared with the frozen triple accepted by RH-70, and that difference must be inserted into the augmented Hardy certificate.

This paper is finite-scale. It does not prove a uniform small-noise peripheral theorem or a continuum spectral limit. It does not close Stage A1 or unconditional Stage A4. It does not construct a self-adjoint Hilbert–Polya operator, prove a $T \log T$ law or prime-power trace formula, locate zeta zeros, or prove the Riemann Hypothesis.

10 Conclusion

The nonnormal spectral factors of the repaired production matrices can be certified with compact, independently checkable ledgers. A stationary Neumann solve, a bordered Newton ball, a perturbed bordered left solve, and a fixed-radius Grushin count together turn the numerical Perron/parity split into a theorem about exact dyadic matrices. The largest complete rank-two error at the archived scales remains below 10^{-13} .

The finite-scale route therefore advances one gate upstream: RH-74 should validate normalized source and observation construction and execute the full upstream-to-frozen Hardy bridge. Whether these constants remain controlled uniformly as $\sigma \downarrow 0$ is deliberately left for later papers.

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